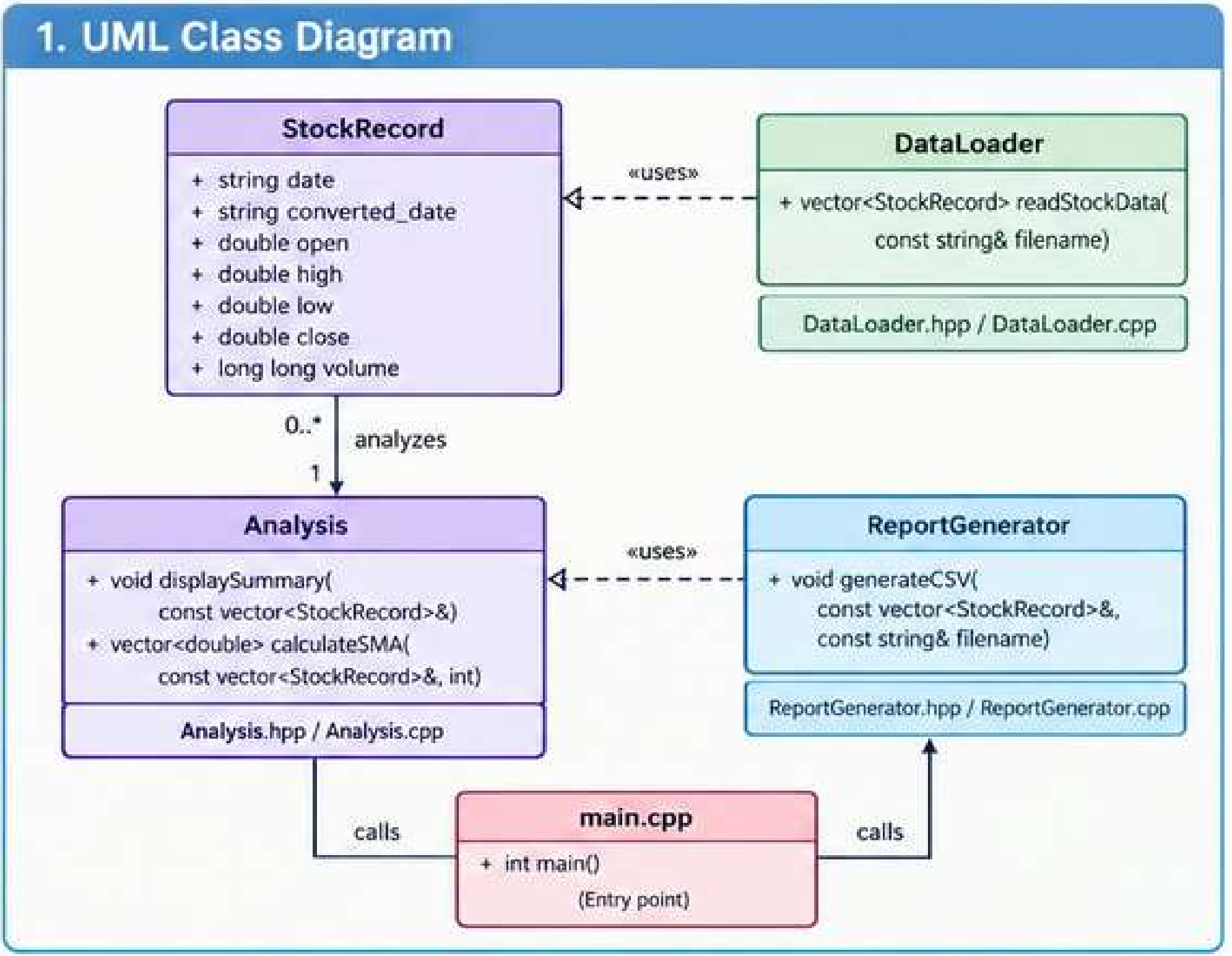
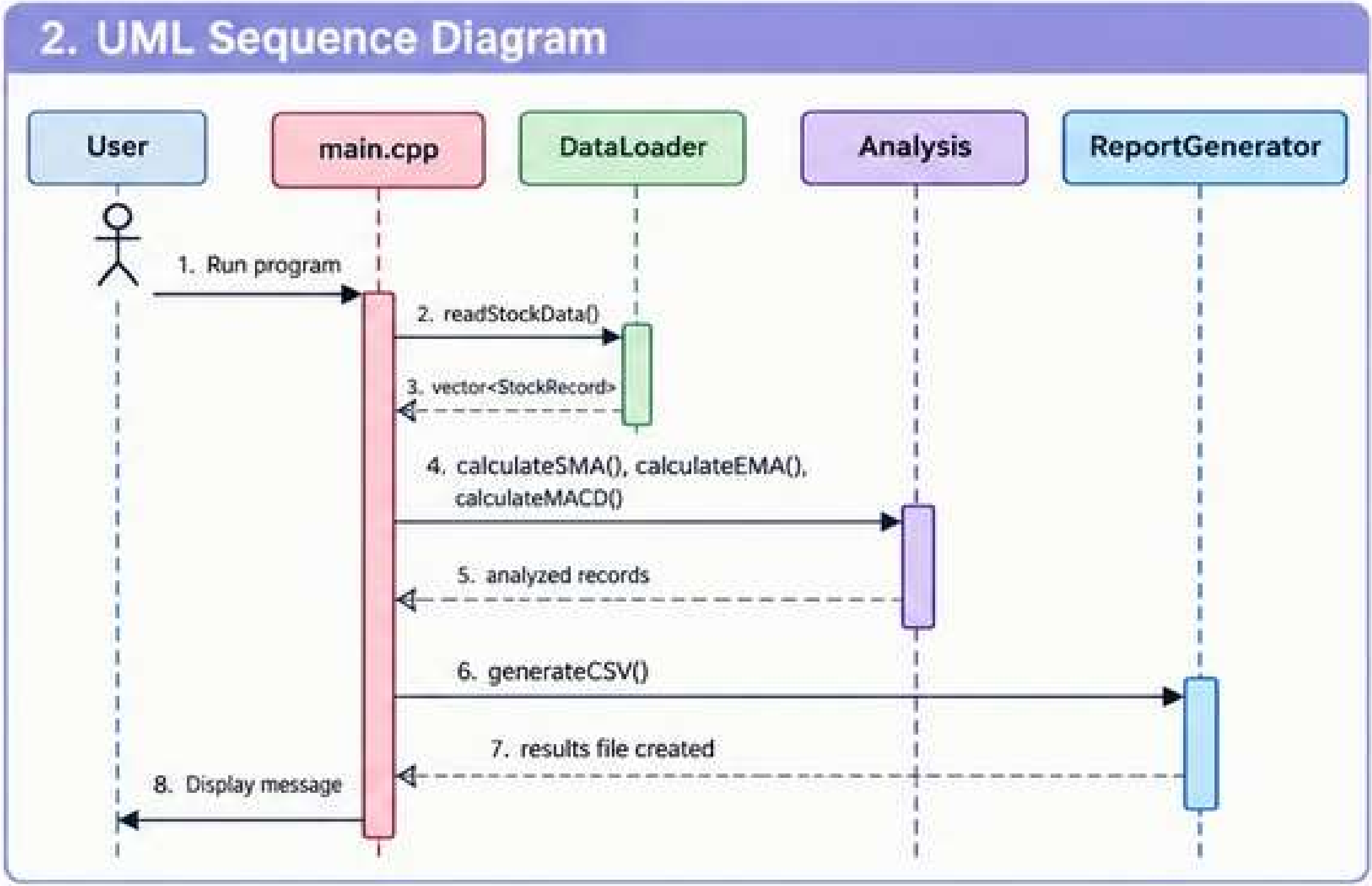
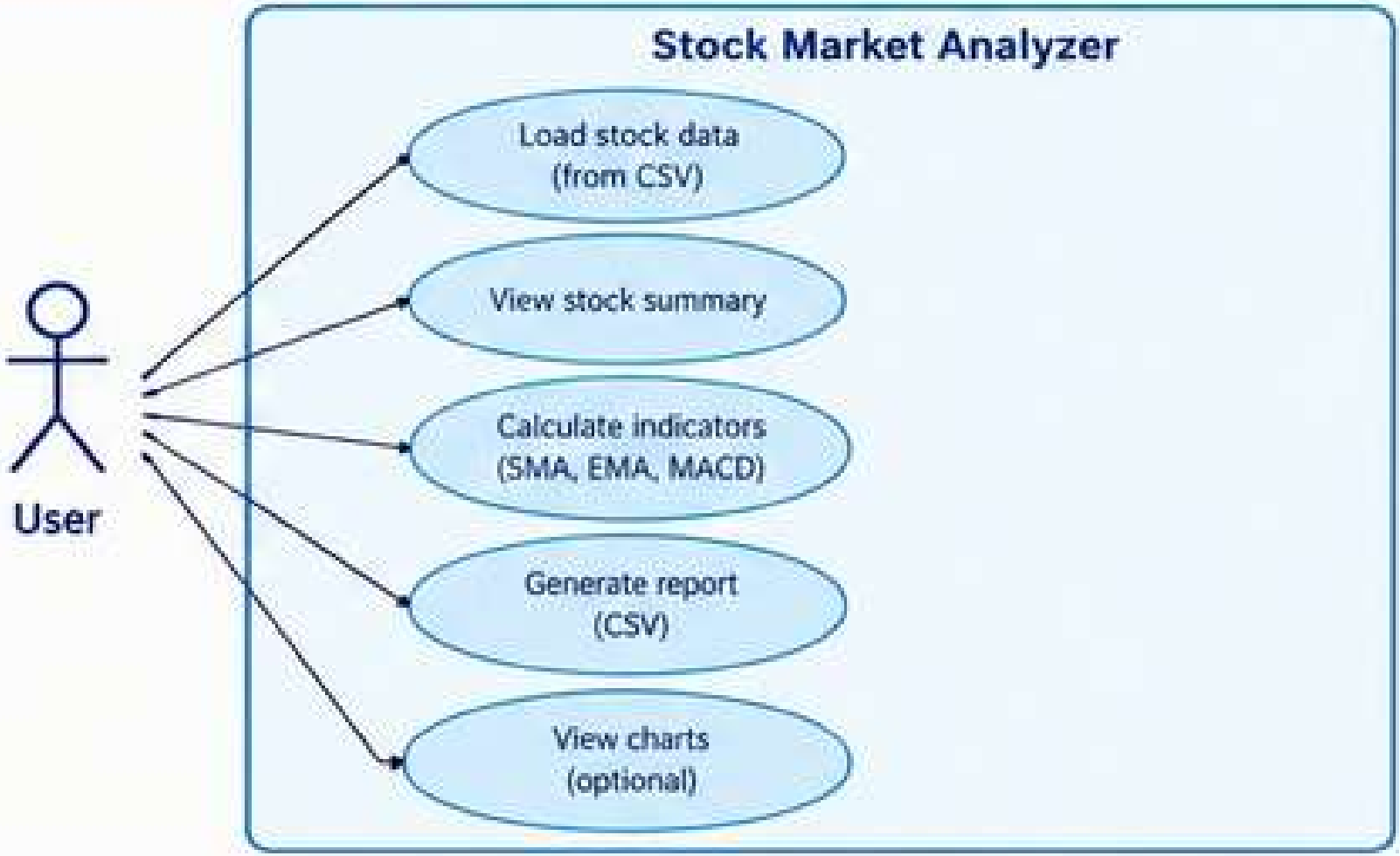
1. UML Class Diagram



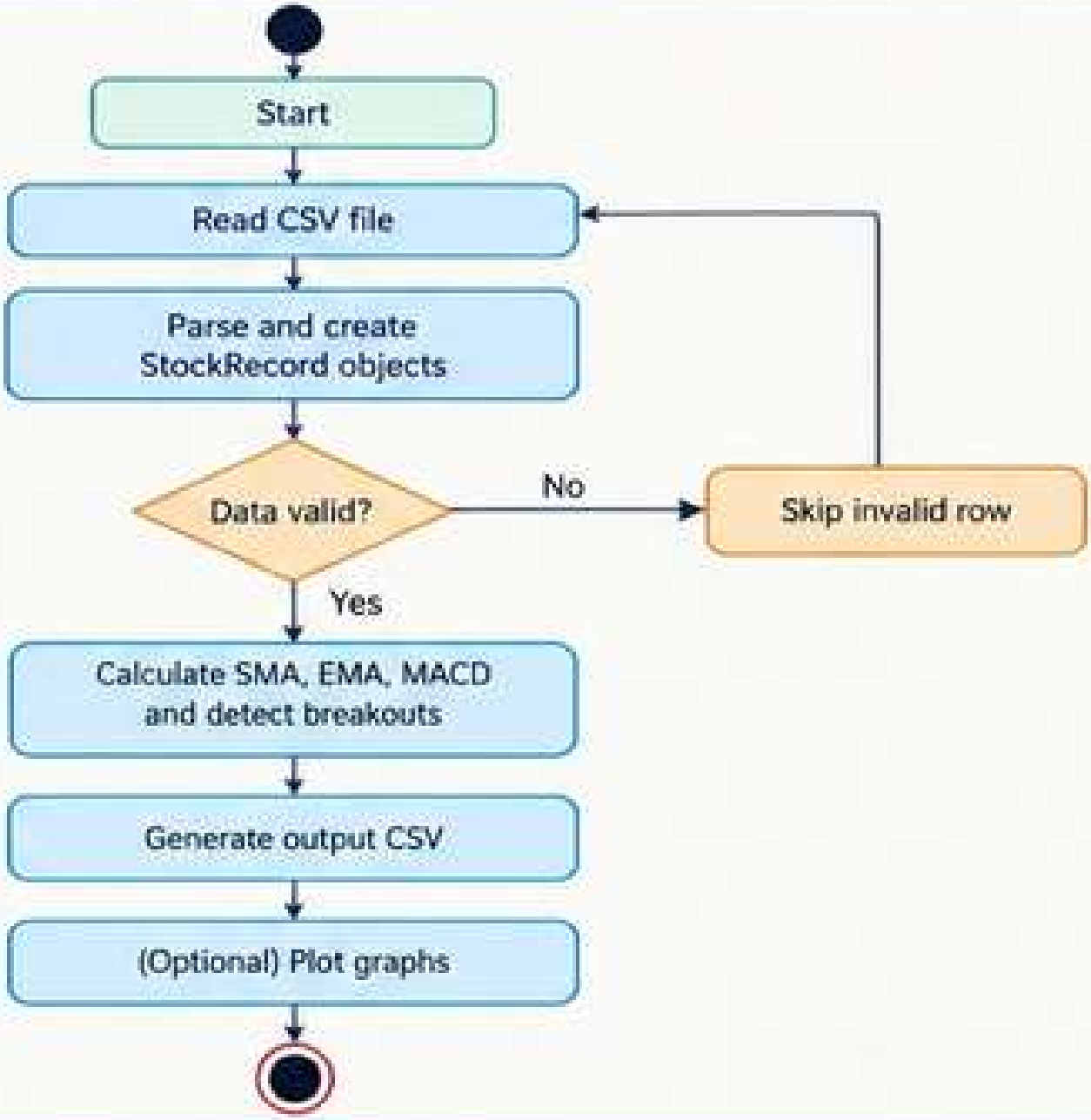
2. UML Sequence Diagram

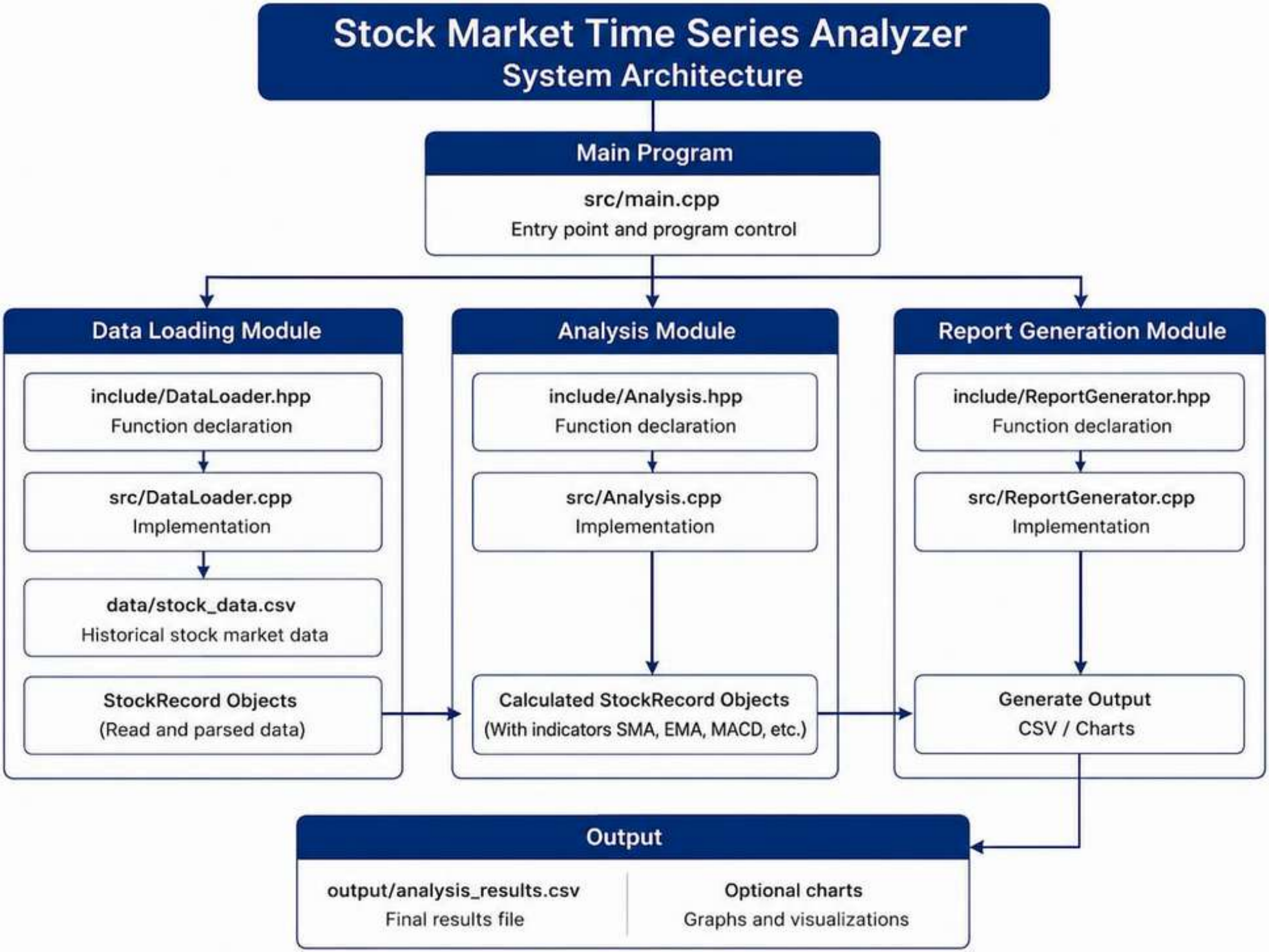


3. UML Use Case Diagram



4. UML Activity Diagram




Stock Market Time Series Analyzer
System Architecture
Main Program
src/main.cpp
Entry point and program control
Data Loading Module
include/DataLoader.hpp
Function declaration
src/DataLoader.cpp
Implementation
data/stock_data.csv
Historical stock market data
StockRecord Objects
(Read and parsed data)
Analysis Module
include/Analysis.hpp
Function declaration
src/Analysis.cpp
Implementation
Calculated StockRecord Objects
(With indicators SMA, EMA, MACD, etc.)
Report Generation Module
include/ReportGenerator.hpp
Function declaration
src/ReportGenerator.cpp
Implementation
Generate Output
CSV / Charts
Output
output/analysis_results.csv
Final results file
Optional charts
Graphs and visualizations